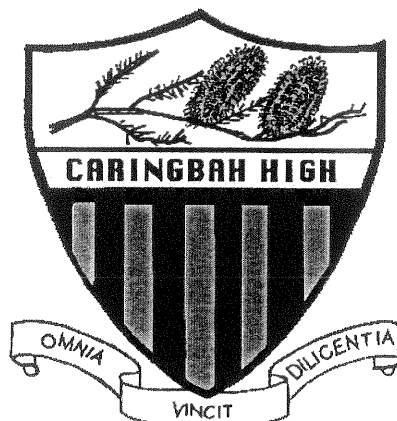


CARINGBAH HIGH SCHOOL



YEAR 12 (2011)

ASSESSMENT TASK # 1 (DEC 2010)

2 UNIT MATHEMATICS

- * Time allowed – 1 hour**
- * Start each question on a new page**
- * Do not use liquid paper**
- * Show all necessary working**
- * Approved calculators permitted**

Question 1 (10 marks)**Marks**

- (a) Using the discriminant determine the number of roots of the quadratic equation $x^2 - 2x - 5 = 0$ (you do not have to find the roots) (1)
- (b) What is the equation of the axis of symmetry of $y = x^2 - 6x$ (1)
- (c) Find the derivative of the following
- (i) $y = \frac{3}{x^2}$
- (ii) $y = \sqrt{2x - 5}$
- (iii) $y = \frac{(x-1)}{(2x-3)}$
- (iv) $f(x) = ax^2 + bx + c$ (6)
- (d) Find the second derivative of $f(x) = (\frac{1}{2}x - 4)^4$ (2)

Question 2 (10 marks)**Marks**

- (a) If $x^2 - 2Kx + 1 = 0$ has real roots find the value(s) of K (2)
- (b) Solve for x the inequality $x^2 - 7x + 12 \leq 0$ (2)
- (c) Find the values of A, B, C if $x^2 + x - 2 \equiv A(x-2)^2 + Bx + C$ (2)
- (d) If $f(x) = x^2 + 5x + 3$ find any value of x for which $f^1(x) = 39$ (2)
- (e) A person has a fever and his temperature is increasing at a decreasing rate. Draw a graph of this with temperature (T) on the vertical axis and time (t) on the horizontal axis. (2)

Question 3 (10 marks)**Marks**

(a) Write the first 3 terms of $T_n = (-1)^n \times \frac{1}{n}$ (1)

(b) Find the common difference in the arithmetic series (1)

$$\sqrt{3} + \sqrt{12} + \sqrt{27} \dots\dots\dots$$

(c) If one root of $3x^2 - 8x + K = 0$ is 3 times the other root find K (3)

(d) If α, β are the roots of $2x^2 + x - 8$ find the value of

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\alpha^2 + \beta^2$ (4)

(e) For the arithmetic series $\{4 + 7 + 10 + \dots\dots\dots\}$

Find the general term T_n in its simplest form (1)

Question 4 (10 marks)**Marks**

(a) For what value of n will one root of the equation

$$(n-2)x^2 + (n+2)x + 2n+1 = 0$$

be the reciprocal of the other. (2)

(b) In an arithmetic series $T_{10} = 5$ and $T_{17} = 54$ Find

(i) the first term (1)

(ii) common difference (1)

(c) Find the seventh term of the geometric series $54 - 18 + 6 \dots\dots$ (2)

Question 4 cont'd**Marks**

(d) An author writes a manuscript so that on the first day he writes 54 pages, on the second day 36 pages and so on each succeeding day he writes $\frac{2}{3}$ rds of the number of pages of the preceding day.

(i) How many pages does he write on the fifth day (1)

(ii) How many pages has he written in 5 days (1)

(iii) What is the maximum number of pages he will write (2)

Question 5 (10 marks)**Marks**

(a) For the curve with equation $y = 2 + 9x - 3x^2 - x^3$

(i) Find any stationary points (2)

(ii) Determine the nature of the stationary points (2)

(iii) Find any inflexion points (2)

(iv) Draw a neat $\frac{1}{2}$ page sketch of this curve including the above information and y intercept. (4)

END OF EXAM

Question 1

(a) $b^2 - 4ac = \Delta$
 $= (2)^2 - 4(1)(-5)$
 $= 24$
 $\therefore \Delta > 0$
 2 roots

(b) $3 = x$

(c) (i) $y' = -\frac{6}{x^3}$

(ii) $y' = \frac{1}{\sqrt{2x-5}}$

(iii) $y' = \frac{-1}{(2x-3)^2}$

(iv) $f'(x) = 2ax + b$

(d) $f''(x) = 3(\frac{1}{2}x - 4)^2$

Question 2

(a) $x^2 - 2Kx + 1 = 0$

now $\Delta \geq 0$

$(-2K)^2 - 4(1)(1) \geq 0$

$4K^2 - 4 \geq 0$

$(K-1)(K+1) \geq 0$

$\therefore K \leq -1 \cup K \geq 1$

(b) $x^2 - 7x + 12 \leq 0$

$(x-3)(x-4) \leq 0$

$\therefore 3 \leq x \leq 4$

(c) $x^2 + x - 2 = A(x^2 - 4x + 4) + Bx + C$

$x^2 + x - 2 = Ax^2 + x(B-4A) + 4A + C$

$\therefore A=1, B-4A=1, 4A+C=-2$
 $B=5 \quad C=-6$

(d) $f(x) = x^2 + 5x + 3$

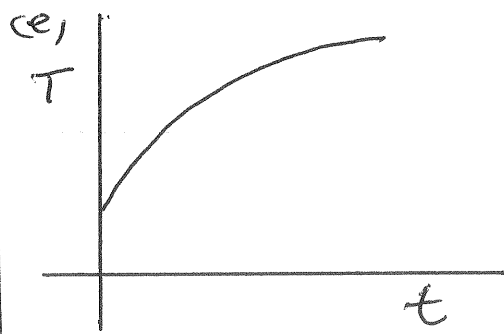
$f'(x) = 2x + 5$

$\therefore 2x + 5 = 39$

$2x = 34$

$x = 17$

Question 2



Question 3

(a) $T_1 = -1$

$T_2 = \frac{1}{2}$

$T_3 = -\frac{1}{3}$

(b) $\sqrt{3}$

(c) $3x^2 - 8x + K = 0$

let roots be

$\alpha, 3\alpha$

$\therefore 4\alpha = \frac{8}{3}$

$\alpha = \frac{2}{3}$

$\therefore 3\alpha^2 = \frac{K}{3}$

$3(\frac{2}{3})^2 = \frac{K}{3}$

$\therefore K = 4$

(d) $2x^2 + x - 8 = 0$

(i) $\alpha + \beta = -\frac{1}{2}$

(ii) $\alpha\beta = -4$

(iii) $\alpha^2 + \beta^2$
 $= (\alpha + \beta)^2 - 2\alpha\beta$
 $= \frac{1}{4} + 8$
 $= 8\frac{1}{4}$

(e) $T_n = 4 + (n-1)3$

$T_n = 3n + 1$

Question 4

(a) Since $\alpha\beta = 1$

$\frac{2n+1}{n-2} = 1$

$2n+1 = n-2$

$n = -3$

(b) $5 = a + 9d \quad \text{--- (1)}$

$54 = a + 16d \quad \text{--- (2)}$

(2) - (1)

$49 = 7d$

$d = 7$

$a = -58$

(c) $T_n = ar^{n-1}$

$T_7 = 54(-\frac{1}{3})^6$

$T_7 = \frac{2}{27}$

(d) $54, 36, 24, \dots$

(i) $T_5 = 54(\frac{2}{3})^4$

$= 10\frac{2}{3}$

(ii) $S_5 = 54 \left[\frac{1 - (\frac{2}{3})^5}{1 - \frac{2}{3}} \right]$

$S_5 = 140\frac{2}{3}$

(iii) $S_\infty = \frac{a}{1-r}$

$= \frac{54}{\frac{1}{3}}$

$= 162$

Question 5

$$a_1 y = 2 + 9x - 3x^2 - x^3$$

$$y' = 9 - 6x - 3x^2$$

$$\text{when } y' = 0$$

$$9 - 6x - 3x^2 = 0$$

$$(i) \quad x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$\begin{bmatrix} x = -3 \\ y = -25 \end{bmatrix} \quad \begin{bmatrix} x = 1 \\ y = 7 \end{bmatrix}$$

$$ii) \quad y'' = -6 - 6x$$

$$\text{when } x = -3 \quad \text{when } x = 1$$

$$y'' > 0$$

$$y'' < 0$$

$\therefore$ Min T.P.

(iv)

ii) Inflection points occur
when $y'' = 0$

$$-6 - 6x = 0$$

$$6x = -6$$

$$x = -1$$

$$[-1, -9]$$

